

A Unified Vision-Language Model for PSMA PET/CT Report Generation, Visual Question Answering, and Lesion Segmentation

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Abstract

Accurate PSMA PET/CT interpretation is central to prostate cancer management, yet existing PET/CT AI models typically address isolated tasks. We propose a unified PSMA PET/CT vision-language model for report generation, visual question answering, and lesion segmentation. The framework adopts an LLaVA-style architecture, comprising a PET/CT vision encoder, an MLP-Mixer projection module, a LoRA-tuned large language model, and a 3D segmentation branch. Training followed a four-stage strategy: vision encoder pretraining, projection-layer alignment, VLM fine-tuning, and final multitask tuning. Language tasks used 5,747 PSMA PET/CT datasets with paired reports, while segmentation used the PSMA subset of AutoPET. The model outperformed PET2REP and a CT-based baseline across standard report-generation metrics, improved performance across VQA question types, and achieved higher Dice and lesion-level overlap F1 than SegAnyPET and nnUNet. These results support the feasibility of a unified

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framework for structured, interactive, interpretable PSMA PET/CT analysis with voxel-level grounding within a single multitask model architecture.

Keywords: Deep learning; PSMA PET; vision-language model; segmentation; report generation

1 Introduction

Following regulatory approval and rapid clinical adoption, prostate-specific membrane antigen (PSMA) PET has become an essential tool for prostate cancer (PCa) management. It outperforms CT and bone scanning in detecting pelvic lymph nodes and distant metastases^{1;2}, and may also facilitate intraprostatic tumor localization and contouring^{3;4}. For PCa biochemical recurrence monitoring, PSMA PET has been demonstrated to be substantially more sensitive than other imaging modalities, especially for prostate-specific antigen (PSA) levels < 1 ng/mL^{5–7}. PSMA PET is now routinely used for baseline staging, detection of biochemical recurrence, and restaging after therapy. Additionally, quantitative PSMA PET imaging plays a central role in treatment response and therapy eligibility for targeted radioligand therapies such as ¹⁷⁷Lu-PSMA^{8;9}. Consensus guidelines^{10–12} increasingly emphasize PSMA PET-derived whole-body tumor burden^{13;14}, lesion distribution¹⁵, and response patterns¹⁶ as key determinants of patient outcomes and suitability for targeted radioligand therapies. Therefore, PSMA PET/CT interpretation requires not only visual lesion detection, but also reproducible lesion localization, quantitative uptake assessment, and whole-body disease burden estimation.

Clinical interpretation and reporting of PET imaging remain largely manual and highly dependent on reader expertise. This limitation is particularly evident in PSMA PET imaging, which has been rapidly adopted in clinical practice, yet standardized reporting remains inconsistently applied. Although professional societies have advocated for structured PSMA PET reporting frameworks, a recent study reported that only 11% of clinical PSMA PET reports adhered to a standardized format, highlighting substantial heterogeneity and inconsistency in real-world practice¹⁷. Such variability may contribute to misinterpretation, inconsistent follow-up or work-up recommendations, and delays in patient management. These challenges motivate the development of automated frameworks for PSMA PET interpretation and reporting. Vision-language models (VLMs), building on architectures such as CLIP¹⁸ and LLaVA¹⁹, provide a powerful framework for integrating visual

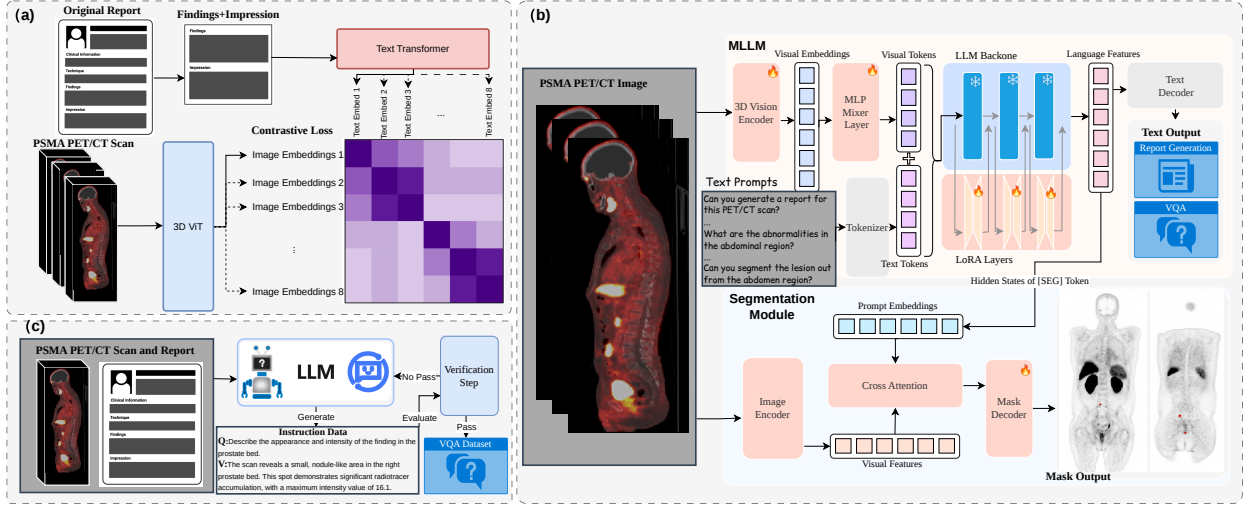


Figure 1: Illustration of the proposed model architecture: (a) The architecture of the PET/CT vision encoder. Contrastive loss was used to train the vision and text encoders to pair radiology reports with the corresponding PET/CT scans. (b) An overview of the proposed methods, which consists of a LLaVA-like MLLM and an nnUNet-based segmentation module, connected by passing the hidden states of [SEG] Token from the MLLM to the segmentation model. (c) The curation process of generating VQA pairs using Gemini-2.5-pro. Details of each module are given in Sec. 4.

and textual information and have been widely applied to medical image analysis, including classification²⁰, image-text retrieval²¹, report generation²², visual question answering (VQA)^{23;24}, and interactive annotation²⁵. Compared with the growing number of CT- and MR-oriented medical VLMs, PET-specific VLMs remain relatively underexplored. Existing PET VLMs^{26–28} have shown potential for image-language alignment and instruction-following, but they have not been designed to jointly support PSMA PET/CT clinical language tasks and 3D lesion segmentation within a single framework.

In this study, we develop a unified PSMA PET/CT VLM for report generation, VQA, semantic lesion segmentation, and language-guided lesion segmentation. The proposed framework combines clinical language tasks and voxel-level lesion segmentation within a single instruction-driven architecture, allowing the model to generate text responses and produce lesion masks according to task-specific prompts. To enable language-guided segmentation, we adapt a LISA-like²⁹ language-guided segmentation interface to the VLM framework and replace the generic segmentation decoder with an nnUNet-style 3D lesion segmentation module. Because anatomical context is important for distinguishing physiologic PSMA uptake from suspicious lesions, we incorporate TotalSegmentator-

derived organ masks as auxiliary anatomical supervision. The model is trained using image-report pairs for report generation, report-derived question-answer pairs for VQA, and lesion-mask annotations for voxel-level segmentation. Through this design, the proposed framework supports both clinical text generation and explicit lesion segmentation for whole-body PSMA PET/CT imaging. Fig. 1 illustrates the architecture of the proposed unified PSMA PET/CT VLM.

2 Results

For image-text retrieval, report generation and VQA tasks, we included 5,747 whole-body PSMA PET/CT image-report paired studies acquired with tracer 18F-DCFPyL between 2022 and 2025. For lesion segmentation task, we utilized PSMA subset from the autoPET challenge³⁰, containing 597 whole-body PSMA PET/CT image-mask paired studies from 378 male patients acquired at LMU University Hospital Munich between 2014 and 2022.

2.1 PET/CT image-text retrieval

Image-to-text and text-to-image retrieval were used to evaluate the image-text alignment learned by the PET/CT vision encoder, referred to as PET/CT-CLIP. In image-to-text retrieval, a PET/CT scan was used as the query to retrieve its corresponding report or text description, and in text-to-image retrieval, a report or text query was used to retrieve the matched PET/CT scan. Retrieval performance was assessed using Recall@K, defined as the percentage of queries for which the matched item was retrieved within the top K results. The proposed PET/CT-CLIP model was compared with random retrieval and CT-CLIP (14), which served as a CT-based vision-language baseline. As shown in Table 1, PET/CT-CLIP achieved the highest Recall@K values across all evaluated retrieval settings. For image-to-text retrieval, PET/CT-CLIP improved Recall@100 from 14.17% with CT-CLIP to 35.57%. For text-to-image retrieval, Recall@100 increased from 8.52% with CT-CLIP to 38.87%. Similar improvements were observed at lower K values, indicating improved alignment between whole-body PET/CT scans and corresponding text descriptions after PET/CT-specific pretraining.

Table 1: Performance comparison for PET/CT image-text retrieval. Recall@ K is reported for image-to-text and text-to-image retrieval using random selection, CT-CLIP, and the proposed PET/CT-CLIP model. Values are reported as mean [95% CI].

Task	Metric	Random	CT-CLIP	PET/CT-CLIP (Proposed)
Image-to-text	Recall@5	0.35 [0.09, 0.70]	1.04 [0.52, 1.74]	3.13 [2.17, 4.17]
	Recall@10	0.78 [0.35, 1.31]	1.83 [1.13, 2.61]	5.13 [3.91, 6.43]
	Recall@50	4.52 [3.30, 5.65]	7.83 [6.26, 9.39]	19.91 [17.65, 22.35]
	Recall@100	7.48 [6.00, 9.04]	14.17 [12.17, 16.17]	35.57 [32.61, 38.26]
Text-to-image	Recall@5	0.87 [0.43, 1.48]	0.43 [0.09, 0.87]	2.43 [1.65, 3.30]
	Recall@10	0.43 [0.09, 0.87]	0.78 [0.26, 1.30]	4.87 [3.74, 6.09]
	Recall@50	5.13 [3.83, 6.35]	4.35 [3.22, 5.65]	23.83 [21.30, 26.17]
	Recall@100	8.26 [6.78, 9.74]	8.52 [6.87, 10.17]	38.87 [36.08, 41.83]

2.2 Report generation

Radiology report generation was evaluated on the held-out PSMA PET/CT test set using ROUGE³¹, BLEU³², METEOR³³, CIDEr³⁴, and BERTScore³⁵. These metrics were used to assess different aspects of report quality, including lexical overlap, phrase-level agreement, and contextual similarity between generated and reference reports. The proposed model was compared with two representative baselines. PET2REP²⁶ was selected because it is a PET/CT-based report generation model and therefore provides the most directly relevant comparison. MedVL-SAM2³⁶ was also included as a CT-only multimodal baseline, given the limited number of publicly available PET-specific report generation models. This comparison was used to assess whether PET/CT-specific visual-language training improves report generation beyond a general 3D CT vision-language framework. All models were evaluated under the same inference setting. Table 2 summarizes the quantitative results, and qualitative examples are shown in Fig. 2. As shown in Table 2, the proposed model achieved the best performance across all evaluated metrics. Compared with PET2REP, the proposed model improved ROUGE from 16.73 to 23.42, BLEU from 27.40 to 36.17, METEOR from 16.98 to 19.62, CIDEr from 0.53 to 3.13, and BERTScore from 84.19 to 86.13. Larger improvements were observed relative to MedVL-SAM2, particularly for BLEU, METEOR, and CIDEr. These results suggest that PET/CT-specific multimodal training improves the generation of reports that are more closely aligned with reference PSMA PET/CT interpretations.

Table 2: Comparison of MedVL-SAM2, PET2REP, and the proposed model using ROUGE, BLEU, METEOR, CIDEr, and BERTScore. Values are reported as mean [95% CI].

Metric	MedVL-SAM2	PET2REP	Proposed Model
ROUGE	12.4605 [12.3097, 12.6281]	16.7270 [16.4854, 16.9981]	23.4173 [22.7848, 24.0766]
BLEU	9.1161 [8.6978, 9.5715]	27.3971 [26.9668, 27.8269]	36.1738 [35.3247, 37.0192]
METEOR	7.6891 [7.5851, 7.7998]	16.9818 [16.7829, 17.1689]	19.6153 [19.2467, 19.9958]
CIDEr	0.0151 [0.0071, 0.0571]	0.5282 [0.3971, 0.6689]	3.1294 [1.8938, 4.6403]
BERTScore	82.6387 [82.5830, 82.7003]	84.1922 [84.1096, 84.2757]	86.1251 [86.0005, 86.2579]

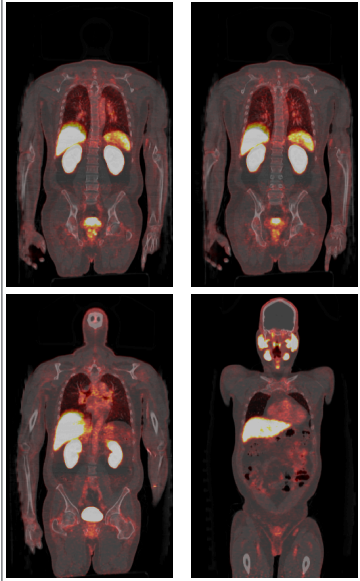
Report Generation			
	<Ground Truth>: Findings: Head and Neck: No focal active or suspicious cervical lymphadenopathy. No focal abnormal activity within the brain which is unremarkable on contrast CT images. Paranasal sinuses and thyroid unremarkable. Likely physiologic activity within lacrimal and salivary glands noted incidentally. Chest: No focal avid or suspicious pulmonary parenchymal abnormalities. No focal avid or suspicious mediastinal or axillary lymphadenopathy. Abdomen and Pelvis: No focal abnormal activity within the liver which is unremarkable on contrast CT images. Gallbladder is visualized. Adrenals, kidneys, pancreas, and spleen are unremarkable on contrast CT images and without focal suspicious activity on corresponding PET images. Nodular activity is seen in region of left greater in the right prostate. For example along lateral mid left prostate with SUVs 8.5 on. Subtle nodular activity in region of mid to superior right prostate with SUV 4.3 on. No focal avid or suspicious abdominal or pelvic lymphadenopathy. Musculoskeletal: No focal avid or suspicious lytic or blastic osseous lesions. Impression: 1. Nodular activity within the prostate, compatible with biopsy-proven prostate carcinoma 2. No suspicious activity to suggest regional nodal or distant metastases.		
	<Proposed Model>: Findings: Head and Neck: No suspicious focal uptake in the brain. No PSMA avid or enlarged cervical lymphadenopathy. Sinuses are overall well aerated. Chest: No evidence of PSMA avid pulmonary mass, pneumothorax, or significant pleural effusion. Left lung base probable granuloma with some additional sites of possible mild scarring versus atelectasis. Scattered atherosclerotic changes including the coronary vessels. The thoracic aorta is within upper limits of normal in caliber. Abdomen and Pelvis: Focal uptake involving the left anterior apex/midline of the prostate gland, SUV 9, consistent with primary prostatic malignancy. No additional suspicious foci of uptake involving the prostate gland. Limited evaluation of urinary bladder segment underdistention and excreted activity is grossly unremarkable. No significant ascites. No evidence of PSMA avid or enlarged dominant all pelvic lymphadenopathy. Physiologic activity involving the liver, nonenlarged spleen, kidneys, and bowel. Some fatty nodularity of the left adrenal gland without suspicious focal uptake probably represents adenomatous changes. The pancreas and gallbladder are grossly unremarkable. Some scattered colonic diverticula without evidence to suggest acute diverticulitis. Musculoskeletal: No evidence of PSMA avid lytic or blastic osseous lesions. Some scattered degenerative changes. Impression: PSMA avid primary prostatic malignancy without evidence suspicious for PSMA avid nodal or distant metastasis.		
	<PET2REP>: FINDINGSHead and Neck - No abnormal radiotracer uptake is visualized within the brain parenchyma. - Expected physiologic activity is seen within the pituitary gland, bilateral lacrimal glands, and salivary glands. - The paranasal sinuses are clear. - There is no evidence of PSMA-avid cervical lymphadenopathy. Chest - The lungs demonstrate no abnormal radiotracer uptake. - There is no pleural effusion and no pericardial effusion. - There is no evidence of PSMA-avid mediastinal, hilar, or axillary lymphadenopathy. - A left upper chest wall cardiac device is present with leads terminating in the right atrium and right ventricle. Abdomen and Pelvis - The prostate is surgically absent with no suspicious PSMA uptake in the resection bed. - The seminal vesicles demonstrate no abnormal PSMA uptake. - There is PSMA-avid retroperitoneal and pelvic lymphadenopathy. - For example: - A left para-aortic lymph node measures approximately 1.3 x 1.3 cm with SUVmax 16.9. - A right perirectal nodal conglomerate measures approximately 3.1 x 1.8 cm with SUVmax 33.4. - Physiologic radiotracer uptake is seen within the liver, spleen, kidneys, and bowel loops. - The gallbladder is present. - The adrenal glands are grossly unremarkable. - The kidneys demonstrate a small left renal cyst. - The bowel demonstrates no evidence of obstruction. Musculoskeletal - The visualized osseous structures demonstrate a focal area of increased PSMA uptake within the right transverse process of the T5 vertebral level with SUVmax approximately 38.1, without a definite CT correlate, concerning for metastatic disease. ##### IMPRESSION - PSMA-avid retroperitoneal and pelvic lymphadenopathy consistent with nodal metastatic prostate carcinoma. - A focal PSMA-avid lesion within the right T5 transverse process is concerning for osseous metastasis. - No evidence of local recurrence in the prostatectomy bed.		

Figure 2: Report-generation examples with color-coded report sections. Representative coronal slices from fused PET/CT images are displayed on the left.

2.3 VQA

VQA was evaluated on the held-out PSMA PET/CT test set using three question formats: description, free-response, and multiple-choice questions. Description and free-response questions were treated as open-ended VQA tasks, in which the model generated text answers based on the PET/CT image and the input question. Multiple-choice questions were treated as closed-ended VQA, in which the model selected one answer from predefined options. These tasks were designed to assess whether the model could interpret clinically relevant PSMA PET/CT findings, including tracer uptake patterns, anatomic localization, disease extent, and impression-level diagnostic information. For open-ended VQA, performance was evaluated using ROUGE, BLEU, METEOR,

Table 3: Comparison of MedVL-SAM2 and the proposed model across VQA tasks, including description, free-response, and multiple-choice questions. Values are reported as mean [95% CI], with confidence intervals estimated using bootstrap resampling.

Task	Metric	MedVL-SAM2	Proposed Model
Description	ROUGE	14.07 [13.88, 14.26]	23.99 [23.62, 24.35]
	BLEU	15.01 [14.51, 15.27]	32.40 [31.55, 32.94]
	METEOR	6.86 [6.75, 6.97]	17.74 [17.48, 17.97]
	CIDEr	1.88 [1.60, 2.22]	18.82 [16.76, 21.11]
	BERTScore	84.86 [84.79, 84.92]	87.45 [87.38, 87.53]
Free response	ROUGE	18.91 [18.52, 19.27]	33.20 [32.50, 33.89]
	BLEU	18.94 [18.32, 19.38]	34.91 [34.30, 35.54]
	METEOR	10.19 [9.96, 10.41]	22.43 [21.89, 22.95]
	CIDEr	17.48 [15.70, 19.27]	91.55 [85.33, 98.04]
	BERTScore	86.90 [86.81, 86.98]	89.46 [89.33, 89.58]
Multiple choice	Accuracy	70.89 [69.29, 72.39]	83.60 [82.39, 84.87]

CIDEr, and BERTScore, and for multiple-choice VQA, accuracy was used as the evaluation metric. Because few PET/CT-specific VQA models are publicly available, MedVL-SAM2 was used as the comparison model. Although MedVL-SAM2 was developed for CT-based multimodal analysis, it provides a relevant 3D medical vision-language baseline for evaluating whether PET/CT-specific training improves multimodal reasoning in this setting. As shown in Table 3, the proposed model outperformed MedVL-SAM2 across all VQA tasks and metrics. For description-based VQA, the proposed model improved ROUGE from 14.07 to 23.99, BLEU from 15.01 to 32.40, METEOR from 6.86 to 17.74, CIDEr from 1.88 to 18.82, and BERTScore from 84.86 to 87.45. For free-response VQA, larger gains were observed, with ROUGE increasing from 18.91 to 33.20 and CIDEr from 17.48 to 91.55. In the multiple-choice setting, accuracy increased from 70.89% to 83.60%. These results indicate that PET/CT-specific multimodal training improves both open-ended answer generation and closed-ended diagnostic question answering. The consistent improvement across description, free-response, and multiple-choice tasks suggests better alignment between 3D PSMA PET/CT findings and clinically oriented text responses, particularly for questions requiring interpretation of lesion distribution and disease status. Qualitative examples are shown in Fig. 3.

2.4 Lesion Segmentation

Lesion segmentation was evaluated on the PSMA PET/CT test set using case-level Dice coefficient and lesion-level F1 score. Dice was used to quantify voxel-level overlap with the reference lesion

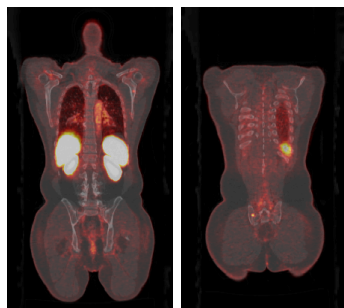
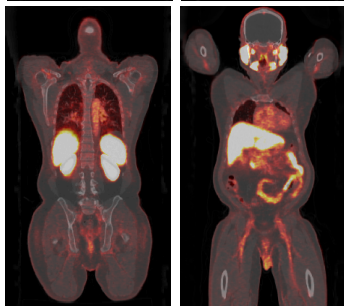
VQA	
Description	
	<Question>: <image>Describe the findings in the chest region, specifically regarding lymphadenopathy, effusions, and pulmonary nodules. <Ground Truth>: In the chest, there is no abnormally enlarged or suspicious PSMA avid lymphadenopathy in the mediastinum, bilateral hila, and axillary regions. No pleural or pericardial effusions are noted. Scarring and atelectatic change are present, but there are no suspicious PSMA avid pulmonary nodules. <Our model>: In the chest, there is no abnormally enlarged or suspicious PSMA avid lymphadenopathy in the mediastinum, bilateral hila, and axillary regions. No pleural or pericardial effusions are noted. Scarring and atelectatic change are present, but no suspicious PSMA avid pulmonary nodules are identified. <MedVL-SAM2>: No lymphadenopathy was observed in the supraclavicular fossa, axilla, and mediastinum in pathological size and appearance. No pericardial effusion was detected. A few nonspecific pulmonary nodules less than 3 mm in diameter were observed in both lungs.
	<Question>: <image>Describe the findings in the head and neck region. <Ground Truth>: In the head and neck, physiologic tracer uptake is noted in the lacrimal glands and salivary glands. There is no abnormal tracer uptake in the brain parenchyma, oropharynx, thyroid glands, or neck muscles. No suspicious neck lymphadenopathy is identified. <Our model>: In the head and neck, there is no suspicious focal tracer uptake in the visualized brain, and no suspicious PSMA-avid cervical nodes. Physiologic radiotracer activity is noted in the bilateral salivary glands and orbits. <MedVL-SAM2>: The trachea and main bronchi are open, and no pathological lymph node was detected in the mediastinum. The heart and mediastinal vascular structures have a natural appearance.
Free-response	
	<Question>: <image>Are there any convincing PSMA avid lesions of concern identified in this PET/CT scan? If not, what is the overall impression regarding PSMA avidity? <Ground Truth>: No convincing PSMA avid lesions of concern are identified. The overall impression is that there are probably benign bone lesions, as detailed in the report. <Our model>: No convincing PSMA avid lesions of concern are identified. The overall impression is that there is no PSMA avid lesion of concern. <MedVL-SAM2>: No, there are no convincing PSMA avid lesions of concern identified in this PET/CT scan. The overall impression is that the patient has a low level of prostate cancer with minimal metastatic lesions.
	<Question>: <image>What is the nature of the PSMA avidity observed in the sclerotic bone lesions mentioned in the report? (a) High avidity (b) Mild avidity (c) No avidity (d) Moderate avidity <Ground Truth>: (b) Mild avidity <Our model>: (b) Mild avidity <MedVL-SAM2>: (c) No avidity <Question>: <image>Which of the following statements is true regarding the prostatectomy bed based on the report? (a) It shows abnormal tracer uptake. (b) It shows suspicious PSMA avidity. (c) It shows no abnormal uptake. (d) It is not mentioned in the report. <Ground Truth>: (c) It shows no abnormal uptake. <Our model>: (c) It shows no abnormal uptake. <MedVL-SAM2>: (c) It shows no abnormal uptake.

Figure 3: Qualitative VQA results for 3 task types: description, free-response, and multiple-choice questions. Representative coronal slices from fused PET/CT images are displayed on the left, with the corresponding question, model response, and reference answer shown on the right.

masks, and F1 assessed clinically relevant detection errors. F1 was computed under three different settings: (i) "Any overlap", where the prediction lesion was considered true positive if it overlaps with the ground truth target; (ii) "SUVmax match", where the prediction lesion was considered true positive if it overlaps the SUVmax voxel of the reference lesion; (iii) "Dice>0.5", where the prediction lesion was considered true positive if the lesion-level Dice score was greater than 0.5. A predicted lesion that failed to fit the criteria was considered a false positive, and an unmatched ground truth lesion was considered a false negative. Bootstrap random sampling with replacement was used to calculate 95% confidence intervals for each metric. The proposed model was evaluated under two prompting settings: semantic segmentation, in which the instruction explicitly requested segmentation of PSMA-avid lesions, and language-guided segmentation, in which the model was guided by a natural language-based prompt. We compared the proposed model with SegAnyPET³⁷ and nnUNet. SegAnyPET was included as a promptable PET segmentation baseline, whereas nnUNet was used as a strong supervised 3D medical segmentation baseline for lesion segmentation. This

Table 4: Comparison of SegAnyPET, nnUNet, and the proposed model for lesion segmentation. Values are reported as mean [95% CI].

Model	Dice	F1 SUVmax	F1 Overlap	F1 Dice > 0.5
SegAnyPET	0.4653 [0.4327, 0.4980]	0.6517 [0.6126, 0.6933]	0.6648 [0.6278, 0.7070]	0.2844 [0.2572, 0.3132]
nnUNet	0.5498 [0.4962, 0.6012]	0.6369 [0.5911, 0.6780]	0.6526 [0.6068, 0.6924]	0.4923 [0.4393, 0.5416]
Proposed (Semantic)	0.5858 [0.5399, 0.6293]	0.7269 [0.6876, 0.7596]	0.7354 [0.6971, 0.7694]	0.5817 [0.5307, 0.6282]
Proposed (Language-guided)	0.5875 [0.5426, 0.6314]	0.7257 [0.6861, 0.7604]	0.7356 [0.6961, 0.7701]	0.5925 [0.5451, 0.6371]

comparison allowed us to assess whether the proposed multimodal framework could preserve the segmentation strength of a task-specific model while enabling language-guided lesion localization. Quantitative results are summarized in Table 4. The proposed model achieved the highest Dice scores in both semantic and language-guided settings, with Dice values of 0.5858 and 0.5875, respectively, outperforming SegAnyPET (0.4653) and nnUNet (0.5498). Similar improvements were observed for lesion-level detection. Under the "Any overlap" criterion, the proposed model achieved F1 scores of 0.7354 and 0.7356 for semantic and language-guided segmentation, respectively. Under the "SUVmax match" criterion, the corresponding F1 scores were 0.7269 and 0.7257. Using the stricter criterion "Dice>0.5", the proposed model achieved F1 scores of 0.5817 and 0.5925, outperforming both SegAnyPET and nnUNet. The semantic and language-guided settings showed comparable Dice performance, suggesting that the model can maintain lesion segmentation accuracy when guided by natural-language prompts. Qualitative examples in Fig. 4 further illustrate the ability of the proposed model to localize PSMA-avid lesions compared with baseline methods.

3 Discussion

This study presents a unified PSMA PET/CT VLM for report generation, VQA, and instruction-conditioned lesion segmentation. The results suggest that PET/CT-specific multimodal training improves performance on clinically oriented language tasks, including report generation and VQA, while preserving 3D lesion localization via an nnUNet-style segmentation module. The primary contribution of this framework is the development of a single instruction-driven system that supports both clinical language tasks and voxel-level lesion segmentation for PSMA PET/CT analysis. The proposed framework can be clinically useful in several ways. First, report generation can pro-

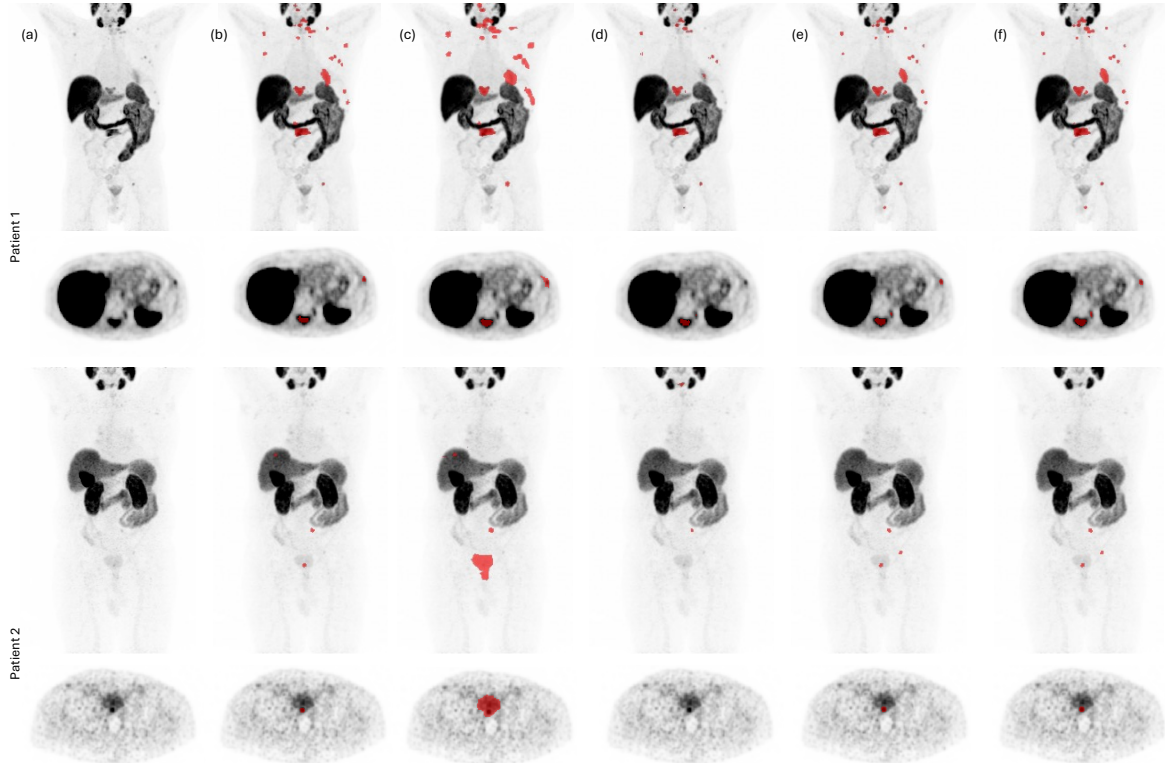


Figure 4: Qualitative visualization of PSMA-avid lesion segmentation. Representative maximum-intensity-projection PET images and axial slices are shown for 2 different cases. (A) Original image, (B) reference lesion mask, (C) SegAnyPET, (D) nnUNet, (E) proposed model with semantic segmentation, and (F) proposed model with language-guided segmentation. Red overlays indicate segmented lesion regions.

vide structured preliminary drafts that summarize major PSMA PET/CT findings. Second, VQA enables interactive querying of disease presence, distribution, and lesion-related findings. Third, lesion segmentation provides 3D voxel-level output that supports semantic and interactive tumor localization, as well as quantitative assessment.

Compared with existing PET-oriented VLMs, which have primarily focused on image-text alignment, report generation, or instruction-following tasks^{26–28}, the proposed framework extends language-based PSMA PET/CT interpretation to include explicit 3D lesion segmentation. Improvements in the language tasks suggest that PSMA-specific vision–language training can better capture the joint PET and CT information than models trained predominantly on other imaging domains. This benefit of domain-specific training is consistent with the nature of PSMA PET/CT interpretation, because identifying clinically meaningful findings and formulating diagnostic im-

pressions require tracer-specific knowledge of physiologic biodistribution, disease-related uptake patterns, and their anatomical correlates on CT. In contrast to dedicated segmentation methods such as nnUNet and SegAnyPET, which directly optimize voxel-level predictions, the proposed framework provides an instruction-conditioned interface: the language model specifies the target through the [SEG] representation, while an nnUNet-style decoder performs dense volumetric prediction. Auxiliary TotalSegmentator-derived organ masks further provide anatomical context that may help distinguish lesions from physiologic uptake in organs with characteristic PSMA expression. The segmentation results therefore support the feasibility of coupling language-conditioned interaction with a dedicated 3D segmentation architecture. Taken together, the report-generation, VQA, and segmentation experiments evaluate complementary aspects of PSMA PET/CT analysis. Report generation assesses the ability to summarize whole-body findings in clinically meaningful text, VQA evaluates the ability to respond to targeted questions about the examination, and segmentation evaluates voxel-level localization of PSMA-avid lesions. Integrating these capabilities within one architecture provides a common instruction-driven interface while preserving the distinct output required by each task.

Several limitations exist in this preliminary study. First, the report-generation and lesion-segmentation datasets were not paired at the case level, which limited direct supervision and evaluation of sentence-level lesion grounding. Specifically, a lesion identified by the segmentation branch was not explicitly constrained to be mentioned in the generated report, and a lesion described in the generated report was not explicitly required to correspond to a predicted mask. Paired image-report-mask datasets will be essential for training and evaluating such lesion-level correspondence. In our future work, we will construct paired image-report-mask datasets to enable explicit training and evaluation of lesion-level correspondence. Second, the VQA dataset was derived from clinical reports and may therefore inherit report-level omissions or interpretation bias. Third, the AutoPET PSMA dataset differed from the in-house cohort in institution, acquisition protocol, and tracer distribution, which may affect segmentation generalizability. Future work will focus on constructing paired PSMA PET/CT datasets that include clinical reports, lesion masks, lesion-level attributes, and prompt-specific annotations from the same examinations. Such datasets would enable direct supervision and more rigorous evaluation of the generated report. In addition, future work will include detailed error analysis and prospective reader studies to determine whether the proposed

unified framework can improve reporting consistency, diagnostic support, and clinical workflow efficiency.

In summary, we developed a unified PSMA PET/CT VLM for report generation, VQA, and lesion segmentation in both semantic and language-conditioned settings. PET/CT-specific vision-language training improved report-generation and VQA performance compared with baseline models, while the 3D segmentation module enabled voxel-level localization of PSMA-avid lesions through both semantic and language-guided prompts. These findings demonstrate the feasibility of a unified framework for clinical language tasks and 3D lesion segmentation, which may support more structured, interactive, and interpretable PSMA PET/CT analysis.

4 Methods

4.1 Dataset

4.1.1 PSMA PET/CT datasets with radiology reports

We retrospectively collected 5,747 whole-body PSMA PET/CT examinations acquired with tracer ^{18}F -DCFPyL between 2022 and 2025. This retrospective study was approved by the University of Florida Institutional Review Board with a waiver of informed consent. For spatial standardization, PET and CT images were resampled to a uniform spacing of $2.73 \times 2.73 \times 2.8$ mm. After resampling, all examinations were cropped to a standardized anatomical field of view ending at the upper thighs and then padded or cropped to a fixed matrix size of $256 \times 256 \times 400$. PET intensities were converted to standardized uptake values (SUV) units. The corresponding clinical reports were converted into structured JSON files. Each JSON record included the study date, findings section, and impression section. This structure preserved the clinically relevant organization of the original reports while enabling paired image-text training and downstream evaluation.

4.1.2 VQA data construction

We constructed the VQA dataset from the paired PSMA PET/CT examinations and structured radiology reports, following the report-derived VQA generation strategy used in CT-CHAT²². For each case, the de-identified findings and impression sections were used as the source text

for question-answer generation. To reflect clinically relevant PSMA PET/CT interpretation, the generated questions focused on disease presence, disease distribution, local recurrence or prostate bed involvement, pelvic and extrapelvic nodal disease, and metastases. For each case, we generated three free-response questions, three multiple-choice questions, and three description-style questions using the Gemini-2.5-Pro API³⁸. Free-response questions were designed to extract concise diagnostic answers from the reports. Multiple-choice questions included one correct answer and plausible distractors. Description questions prompted short summaries of the major PET/CT findings or disease distribution. To improve data quality, VQA generation was performed using a two-step iterative procedure. In the first step, the language model generated candidate question-answer pairs from the structured report. In the second step, the model was prompted to verify whether each answer was directly supported by evidence in the report and to identify the corresponding supporting text. Question-answer pairs were retained only if they were correctly formatted, internally consistent, and supported by explicit report evidence. Unsupported, ambiguous, or malformed entries were discarded or regenerated. The pipeline of the data curation process is shown in Fig. 1. C. The dataset was split at the case level to prevent information leakage between training, validation, and testing sets. After quality control, the dataset comprised 4,023 training cases, 574 validation cases, and 1,150 testing cases. The training set contained 12,033 description, 11,772 multiple-choice, and 12,033 free-response questions; the validation set contained 1,722 questions for each VQA type; and the testing set contained 3,442 description, 3,390 multiple-choice, and 3,442 free-response questions.

4.1.3 AutoPET PSMA lesion segmentation dataset

The segmentation component was developed using the PSMA subset from the autoPET challenge³⁰. This dataset contains 597 whole-body PSMA PET/CT studies from 378 male patients with suspected or diagnosed prostate cancer, acquired at LMU University Hospital Munich between 2014 and 2022. The lesion label was defined as a binary voxel-level mask of PSMA-avid tumor lesions, manually annotated in 3D on PET images. All PET/CT images and lesion masks were processed using the same spatial normalization pipeline as the in-house PSMA dataset. This dataset was used for semantic tumor segmentation training and quantitative evaluation of lesion-level segmentation performance. The dataset was split at the patient level into training, validation, and testing sets of 419, 59, and 119 studies, respectively.

4.1.4 TotalSegmentator-derived organ masks

To provide auxiliary anatomical supervision, physiologic uptake-related organ masks were generated from the CT images using TotalSegmentator³⁹. Ten organ or anatomic structure groups were selected based on their common physiologic or tracer-dependent uptake patterns, which may confound lesion detection on PSMA PET/CT. These included the kidneys, urinary bladder, liver, spleen, brain, lungs, heart, prostate, stomach, and salivary glands. For the salivary glands, parotid and submandibular gland masks were grouped into a single salivary gland category. For bilateral structures, left and right masks were merged into one organ class unless otherwise specified. These masks were used as auxiliary pseudo-labels for organ-aware supervision, helping the segmentation branch detect normal biodistribution patterns and distinguish suspicious lesions from physiologic uptake.

4.2 Network architecture

4.2.1 Overview

3D medical vision-language tasks cover a broad spectrum, including report generation, VQA, and segmentation. To support all these tasks within a single model, we introduce a unified formulation that accommodates both text-centric and mask-centric predictions. Given a 3D PET/CT volume $\mathcal{I}_{in} \in \mathbb{R}^{2 \times X \times Y \times Z}$ with dimension of $X \times Y \times Z$ and a sequence of text tokens $\mathcal{T}_{in} \in \mathbb{R}^{N \times D}$ with size of $N \times D$, the unified multi-task problem can be formulated as

$$(\mathcal{T}_{out}, \mathcal{M}_{out}) = \mathcal{F}(\Theta; \mathcal{I}_{in}, \mathcal{T}_{in}), \quad (1)$$

where $\mathcal{F}(\cdot)$ denotes the unified 3D medical VLM, $\Theta = \{\theta_{vlb}, \theta_{seg}\}$ denotes the trainable parameters including the vision-language backbone parameters θ_{vlb} and the segmentation module parameters θ_{seg} , $\mathcal{T}_{out}$ represents the output text tokens, and $\mathcal{M}_{out}$ is the output segmentation mask. The output text tokens $\mathcal{T}_{out}$ are decoded by the tokenizer to generate the text output.

The proposed model adopts a LLaVA-style multimodal framework to support PSMA PET/CT report generation, VQA, and lesion segmentation within a unified instruction-following architecture. The network consists of four main components: a 3D PET/CT image encoder, a multimodal

projection layer, a large language model (LLM), and a language-conditioned nnUNet-style 3D segmentation module. The overall architecture is shown in Fig. 1 (B). The image-language branch encodes 3D PET/CT scans into visual tokens and aligns them with the language model embedding space. The LLM then generates free-text outputs, including radiology reports and answers to diagnostic questions. For segmentation tasks, a special [SEG] token is used to condition voxel-level mask prediction on the segmentation instruction. The final hidden-state representation of this token is projected into the segmentation feature space and fused with PET/CT features through a cross-attention module. This design allows the segmentation output to be conditioned on the text instruction while preserving the strong 3D localization capability of an nnUNet-style encoder-decoder network.

4.2.2 Vision-language alignment

Given an input PET/CT image, a 3D CLIP-based vision encoder E_{vc} is used to extract image features v_{in} , denoted as $v_{in} = E_{vc}(\mathcal{I}_{in}) \in \mathbb{R}^{n \times d}$, where n is the number of visual tokens and d is the token dimension. PET and CT images are provided as two channels to provide both molecular uptake information and anatomical context. Directly passing all 3D visual tokens into the language model is computationally inefficient because 3D patch embedding produces substantially longer token sequences than conventional 2D image inputs. To reduce this burden while preserving spatial information, an adaptive MLP-Mixer projection module⁴⁰ is used to compress the visual tokens and align them with the language model embedding space. Given the visual tokens, the MLP-Mixer projection layer performs both channel and token mixing, producing language-compatible visual tokens that allow the model to reduce the visual token length while maintaining global context. Each MLP-Mixer layer contains two fully connected sublayers with nonlinearities applied independently to each row of the input tensor: the first performs channel mixing, and the second performs token mixing. Omitting layer indices, the mixer layer is formulated as

$$\begin{aligned} U^T &= W_2 \sigma(W_1 \text{LayerNorm}(v_{in}^T)), \\ T_v &= W_4 \sigma(W_3 \text{LayerNorm}(U)), \end{aligned} \tag{2}$$

where σ denotes the GELU activation function, $U \in \mathbb{R}^{\hat{n} \times d}$ and $T_v \in \mathbb{R}^{\hat{n} \times \hat{d}}$ are the visual tokens aligned with the LLM’s text embedding dimensions ($\hat{n} = 512$ and $\hat{d} = 2048$ in our case). The computational complexity of the MLP-Mixer layer is linear with respect to the number of input patches, which enables efficient compression of 3D volumetric embeddings while preserving spatial information. This design is more efficient than directly using a standard linear projection on the full 3D token sequence. The projected visual tokens are concatenated with the text instruction tokens and then passed to the language model, which generates radiology reports, answers clinical questions, and produces task-specific output tokens for segmentation. LoRA layers (rank=128) are applied to the language backbone to enable parameter-efficient fine-tuning while preserving the general language capability of the pretrained model. The overall process is formulated as: $T_{out} = LLM(T_{in} + T_v)$, where T_{in} denotes the tokenized text prompt and T_v denotes the visual tokens produced by the projection layer.

4.2.3 Semantic and language-guided segmentation module

The segmentation branch follows an nnUNet-style 3D encoder-decoder design⁴¹. Unlike the 3D vision encoder, which is optimized for global image-text alignment, the segmentation branch is optimized for dense 3D voxel-level prediction. It receives the preprocessed PET/CT image and extracts multi-resolution features through a residual encoder f_{enc} : $\mathcal{Z} = \{z_1, \dots, z_s\}, z_s \in \mathbb{R}^{C_s \times H_s \times W_s \times D_s}$, where S is the number of scales. These features are then decoded with skip connections to generate the final 3D lesion mask: $y_s = f_{dec,s}(y_{s-1}, z_s)$. For segmentation tasks, the instruction prompt asks the model to identify the target lesion or structure and produce a special [SEG] token. The final-layer hidden representation of this token is extracted from the language model and projected into the segmentation feature space, denoted as $[\text{SEG}]_{hs}$. This representation encodes the target specified by the text instruction, such as PSMA-avid lesion segmentation or a language-guided segmentation request. To condition mask prediction on the text instruction, the projected [SEG] representation is fused with bottleneck segmentation features through a cross-attention module, denoted as $\tilde{z}_s = CA([\text{SEG}]_{hs}, z_s)$, where $[\text{SEG}]_{hs}$ is the query and z_s is the key and value for the cross attention module. This fusion allows the segmentation branch to emphasize image features relevant to the text-specified target before mask decoding. The resulting language-conditioned multi-resolution features are passed to the decoder to produce the final 3D probability map, which

is subsequently converted into a binary lesion mask. This design combines language-conditioned target specification with a specialized medical segmentation backbone.

4.3 Implementation details

The model was trained using a four-stage strategy to progressively establish PET/CT visual encoding, vision-language alignment, task-specific language generation, and voxel-level segmentation. First, the 3D PET/CT vision encoder was trained using contrastive learning¹⁸ to align 3D image features with corresponding report-level text representations. This stage provided an imaging-specific initialization for encoding whole-body PSMA PET/CT images. The training process of the vision encoder is shown in Fig. 1. A. Second, the vision-language projection layer was trained to map PET/CT visual tokens into the language model embedding space. During this stage, the vision encoder and language model were frozen, and only the projection module was optimized using report-generation and VQA samples with the autoregressive language modeling loss. Third, the VLM branch was fine-tuned for report generation and VQA. The PET/CT vision encoder, projection layer, and LoRA parameters of the language model were optimized, while the segmentation branch was not updated. This stage strengthened PET/CT-specific image-text reasoning before introducing voxel-level supervision. Finally, multitask tuning was performed using mixed instructions for report generation, VQA, semantic segmentation, and language-guided segmentation. The VLM branch, segmentation-token interface, cross-attention fusion module, and nnUNet-style segmentation branch were jointly optimized. Text-generation samples were supervised using autoregressive language modeling loss, while segmentation samples were supervised using a combination of cross-entropy and Dice losses. During testing, task-specific prompts were used for report generation, VQA, semantic segmentation, and language-guided segmentation.

5 Statistical analysis

Statistical analysis was performed to quantify the uncertainty and robustness of model performance. Bootstrap resampling was used to estimate 95% confidence intervals for all evaluation metrics. For image-text retrieval, Recall@K was evaluated for both image-to-text and text-to-image retrieval. Report generation was evaluated using ROUGE, BLEU, METEOR, CIDEr, and BERTScore. VQA

performance was evaluated using language-generation metrics for description and free-response questions and accuracy for multiple-choice questions. Lesion segmentation was evaluated using the case-level Dice coefficient and lesion-level F1 scores under the SUVmax-match, any-overlap, and Dice (>0.5) criteria.

6 Data availability

The institutional PSMA PET/CT imaging data and corresponding clinical reports used in this study are not publicly available because of patient privacy and institutional ethical restrictions. De-identified data may be made available to qualified researchers upon reasonable request to the corresponding author, subject to institutional approval and execution of an appropriate data-use agreement. The publicly available AutoPET dataset used for lesion-segmentation experiments can be accessed through its original repository: <https://autopet-iv.grand-challenge.org>.

7 Code availability

Model training was performed using the AdamW optimizer as implemented in PyTorch. For baseline comparisons, we used the CTCLIP model (hf-hub:<https://huggingface.co/datasets/ibrahimhamamci/CT-RATE>) via the HuggingFace hub. All experiments were implemented using PyTorch (v2.7) and Transformers (v4.42), and executed on a cluster equipped with eight NVIDIA B200 GPUs (180 GB each).

Acknowledgements

This work was supported by NIH grants R01EB034692 and R01AG078250.

Author contributions

Y.X. processed the data, developed the code, trained the models, performed the experiments, analyzed the results, prepared the figures, and drafted the manuscript. J.W., Y.Y., W.S., and C.Y. contributed to model and experimental design and interpretation of the results. T.P., Y.Y.,

and Z.Z. collected the data. Y.Z. contributed to the VQA construction pipeline. B.Y. and Z.Z. performed data processing. K.W., S.O., and Y.L. contributed to clinical task design and analysis of the results. K.G. guided the project, provided research support, and advised on technical details and overall direction. All authors reviewed, revised, and approved the manuscript.

Competing interest

The authors declare no competing interests.

References

- [1] Hofman, M. S. *et al.* Prostate-specific membrane antigen PET-CT in patients with high-risk prostate cancer before curative-intent surgery or radiotherapy (proPSMA): a prospective, randomised, multicentre study. *The Lancet* **395**, 1208–1216 (2020). URL <https://linkinghub.elsevier.com/retrieve/pii/S0140673620303147>.
- [2] Hope, T. A. *et al.* Diagnostic Accuracy of ^{68}Ga -PSMA-11 PET for Pelvic Nodal Metastasis Detection Prior to Radical Prostatectomy and Pelvic Lymph Node Dissection: A Multicenter Prospective Phase 3 Imaging Trial. *JAMA Oncology* **7**, 1635 (2021). URL <https://jamanetwork.com/journals/jamaoncology/fullarticle/2783766>.
- [3] Chen, M. *et al.* Combination of ^{68}Ga -PSMA PET/CT and Multiparametric MRI Improves the Detection of Clinically Significant Prostate Cancer: A Lesion-by-Lesion Analysis. *Journal of Nuclear Medicine* **60**, 944–949 (2019). URL <http://jnm.snmjournals.org/lookup/doi/10.2967/jnumed.118.221010>.
- [4] Bettermann, A. S. *et al.* [^{68}Ga]-PSMA-11 PET/CT and multiparametric MRI for gross tumor volume delineation in a slice by slice analysis with whole mount histopathology as a reference standard – Implications for focal radiotherapy planning in primary prostate cancer. *Radiotherapy and Oncology* **141**, 214–219 (2019). URL <https://linkinghub.elsevier.com/retrieve/pii/S0167814019329846>.
- [5] Cornford, P. *et al.* EAU-EANM-ESTRO-ESUR-SIOG Guidelines on Prostate Cancer. Part

- II—2020 Update: Treatment of Relapsing and Metastatic Prostate Cancer. *European Urology* **79**, 263–282 (2021). URL <https://linkinghub.elsevier.com/retrieve/pii/S0302283820307739>.
- [6] Morigi, J. J. *et al.* Prospective Comparison of ^{18}F -Fluoromethylcholine Versus ^{68}Ga -PSMA PET/CT in Prostate Cancer Patients Who Have Rising PSA After Curative Treatment and Are Being Considered for Targeted Therapy. *Journal of Nuclear Medicine* **56**, 1185–1190 (2015). URL <http://jnm.snmjournals.org/lookup/doi/10.2967/jnumed.115.160382>.
- [7] Afshar-Oromieh, A. *et al.* Comparison of PET imaging with a ^{68}Ga -labelled PSMA ligand and ^{18}F -choline-based PET/CT for the diagnosis of recurrent prostate cancer. *European Journal of Nuclear Medicine and Molecular Imaging* **41**, 11–20 (2014). URL <http://link.springer.com/10.1007/s00259-013-2525-5>.
- [8] Eiber, M. *et al.* Evaluation of Hybrid ^{68}Ga -PSMA Ligand PET/CT in 248 Patients with Biochemical Recurrence After Radical Prostatectomy. *Journal of Nuclear Medicine* **56**, 668–674 (2015). URL <http://jnm.snmjournals.org/lookup/doi/10.2967/jnumed.115.154153>.
- [9] Rauscher, I. *et al.* Efficacy, Predictive Factors, and Prediction Nomograms for ^{68}Ga -labeled Prostate-specific Membrane Antigen–ligand Positron-emission Tomography/Computed Tomography in Early Biochemical Recurrent Prostate Cancer After Radical Prostatectomy. *European Urology* **73**, 656–661 (2018). URL <https://linkinghub.elsevier.com/retrieve/pii/S0302283818300101>.
- [10] Kratochwil, C. *et al.* Joint EANM/SNMMI procedure guideline for the use of ^{177}Lu -labeled PSMA-targeted radioligand-therapy (^{177}Lu -PSMA-RLT). *European Journal of Nuclear Medicine and Molecular Imaging* **50**, 2830–2845 (2023). URL <https://link.springer.com/10.1007/s00259-023-06255-8>.
- [11] Hope, T. A. *et al.* SNMMI Consensus Statement on Patient Selection and Appropriate Use of ^{177}Lu -PSMA-617 Radionuclide Therapy. *Journal of Nuclear Medicine* **64**, 1417–1423 (2023). URL <http://jnm.snmjournals.org/lookup/doi/10.2967/jnumed.123.265952>.
- [12] Hofman, M. S. *et al.* [^{177}Lu]-PSMA-617 radionuclide treatment in patients with metastatic

- castration-resistant prostate cancer (LuPSMA trial): a single-centre, single-arm, phase 2 study. *The Lancet Oncology* **19**, 825–833 (2018). URL <https://linkinghub.elsevier.com/retrieve/pii/S1470204518301980>.
- [13] Sartor, O. *et al.* Lutetium-177–PSMA-617 for Metastatic Castration-Resistant Prostate Cancer. *New England Journal of Medicine* **385**, 1091–1103 (2021). URL <http://www.nejm.org/doi/10.1056/NEJMoa2107322>.
- [14] Gafita, A. *et al.* Novel framework for treatment response evaluation using PSMA-PET/CT in patients with metastatic castration-resistant prostate cancer (RECIP 1.0): an international multicenter study. *Journal of Nuclear Medicine* jnumed.121.263072 (2022). URL <http://jnm.snmjournals.org/lookup/doi/10.2967/jnumed.121.263072>.
- [15] Eiber, M. *et al.* Prostate Cancer Molecular Imaging Standardized Evaluation (PROMISE): Proposed miTNM Classification for the Interpretation of PSMA-Ligand PET/CT. *Journal of Nuclear Medicine* **59**, 469–478 (2018). URL <http://jnm.snmjournals.org/lookup/doi/10.2967/jnumed.117.198119>.
- [16] Fanti, S., Hadaschik, B. & Herrmann, K. Proposal for Systemic-Therapy Response-Assessment Criteria at the Time of PSMA PET/CT Imaging: The PSMA PET Progression Criteria. *Journal of Nuclear Medicine* **61**, 678–682 (2020). URL <http://jnm.snmjournals.org/lookup/doi/10.2967/jnumed.119.233817>.
- [17] Vargas, H. A. *et al.* PSMA PET/CT Reporting: Real-Life Inconsistencies, Heterogeneity, and Underutilization of Scoring Systems and Interpretation Frameworks. *Radiology* **317**, e250702 (2025). URL <http://pubs.rsna.org/doi/10.1148/radiol.250702>.
- [18] Radford, A. *et al.* Learning Transferable Visual Models From Natural Language Supervision .
- [19] Liu, H., Li, C., Wu, Q. & Lee, Y. J. Visual Instruction Tuning (2023). URL <http://arxiv.org/abs/2304.08485>. ArXiv:2304.08485 [cs.CV].
- [20] Huemann, Z. *et al.* ConTEXTual Net 3D: Vision-language modeling in PET/CT for visual grounding of positive findings. *Journal of Imaging Informatics in Medicine* (2026).

- [21] Zhang, S. *et al.* Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs (2025). URL <https://arxiv.org/abs/2303.00915>. arXiv:2303.00915.
- [22] Hamamci, I. E. *et al.* Generalist foundation models from a multimodal dataset for 3D computed tomography. *Nature Biomedical Engineering* (2026). URL <https://www.nature.com/articles/s41551-025-01599-y>.
- [23] Wu, C. *et al.* Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data. *Nature Communications* **16**, 7866 (2025).
- [24] Bai, F., Du, Y., Huang, T., Meng, M. Q. H. & Zhao, B. M3d: Advancing 3d medical image analysis with multi-modal large language models (2024). URL <https://arxiv.org/abs/2404.00578>. arXiv:2404.00578.
- [25] Zhao, T. *et al.* A foundation model for joint segmentation, detection and recognition of biomedical objects across nine modalities. *Nature methods* **22**, 166–176 (2025).
- [26] Zhang, Y. *et al.* PET2Rep: Towards Vision-Language Model-Driven Automated Radiology Report Generation for Positron Emission Tomography (2025). URL <http://arxiv.org/abs/2508.04062>. ArXiv:2508.04062 [eess.IV].
- [27] Jiao, W. *et al.* Vision-Language Models for Automated 3D PET/CT Report Generation (2025). URL <http://arxiv.org/abs/2511.20145>. ArXiv:2511.20145 [cs.CV].
- [28] Maqbool, D. *et al.* PETAR: Localized Findings Generation with Mask-Aware Vision-Language Modeling for PET Automated Reporting (2025). URL <http://arxiv.org/abs/2510.27680>. ArXiv:2510.27680 [cs.CV].
- [29] Lai, X. *et al.* LISA: Reasoning segmentation via large language model. *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.* 9579–9589 (2024). URL <https://ieeexplore.ieee.org/document/10658574/>.
- [30] Dexl, J. *et al.* The autoPET3 Challenge: Automated Lesion Segmentation in Whole-Body PET/CT $\{x_{2013}\}$ Multitracer Multicenter Generalization (2026). URL <http://arxiv.org/abs/2605.05775>. ArXiv:2605.05775 [cs.CV].

- [31] Lin, C.-Y. ROUGE: A Package for Automatic Evaluation of Summaries .
- [32] Papineni, K., Roukos, S., Ward, T. & Zhu, W.-J. Isabelle, P., Charniak, E. & Lin, D. (eds) *BLEU: A method for automatic evaluation of machine translation*. (eds Isabelle, P., Charniak, E. & Lin, D.) *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, 311–318 (Association for Computational Linguistics, Philadelphia, Pennsylvania, USA, 2002). URL <https://aclanthology.org/P02-1040/>.
- [33] Satanjeev Banerjee & Alon Lavie. METEOR: An Automatic Metric for MT Evaluation with Improved Correlation with Human Judgments. *Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization* (2005).
- [34] Vedantam, R., Lawrence Zitnick, C. & Parikh, D. CIDEr: Consensus-based image description evaluation. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2015).
- [35] Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q. & Artzi, Y. BERTScore: Evaluating Text Generation with BERT (2020). URL <http://arxiv.org/abs/1904.09675>. ArXiv:1904.09675 [cs.CL].
- [36] Xing, Y. *et al.* MedVL-SAM2: A unified 3D medical vision-language model for multimodal reasoning and prompt-driven segmentation (2026). URL <http://arxiv.org/abs/2601.09879>. ArXiv:2601.09879 [cs.CV].
- [37] Zhang, Y. *et al.* SegAnyPET: Universal Promptable Segmentation from Positron Emission Tomography Images .
- [38] Comanici, G. *et al.* Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities (2025). URL <http://arxiv.org/abs/2507.06261>. ArXiv:2507.06261 [cs.CL].
- [39] Wasserthal, J. *et al.* TotalSegmentator: Robust Segmentation of 104 Anatomic Structures in CT Images. *Radiology: Artificial Intelligence* **5**, e230024 (2023). URL <http://pubs.rsna.org/doi/10.1148/ryai.230024>.

- [40] Xin, Y., Ates, G. C., Gong, K. & Shao, W. Med3DVLM: An Efficient Vision-Language Model for 3D Medical Image Analysis (2025). URL <http://arxiv.org/abs/2503.20047>. ArXiv:2503.20047 [cs.CV].
- [41] Isensee, F., Jaeger, P. F., Kohl, S. A. A., Petersen, J. & Maier-Hein, K. H. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods* **18**, 203–211 (2021). URL <https://www.nature.com/articles/s41592-020-01008-z>.